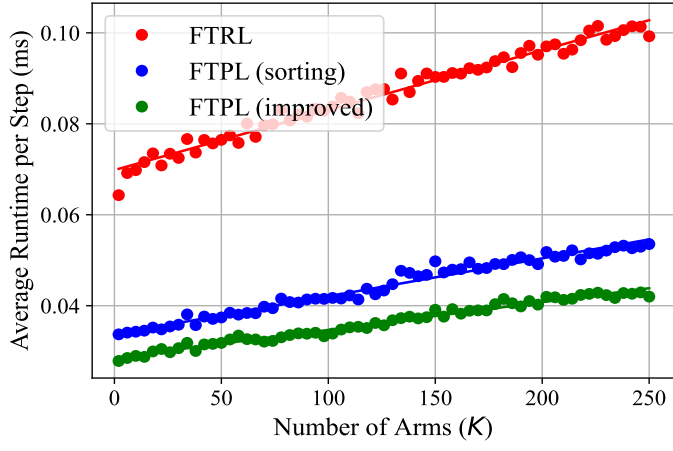
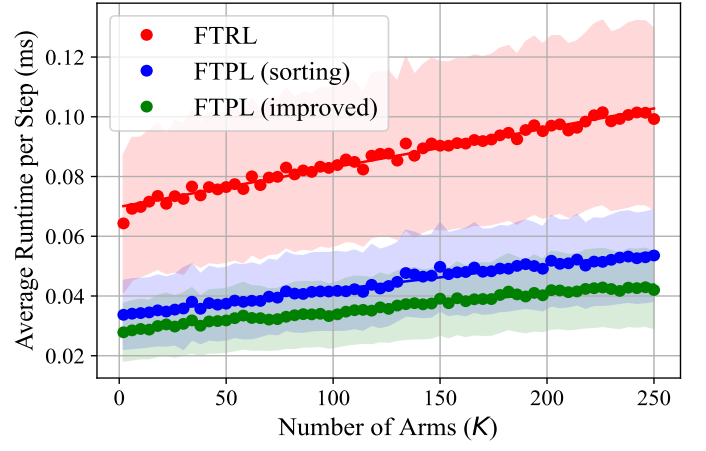




(i) Averaged run time



(ii) Averaged run time with one standard deviation

Figure A: Run time comparison with improved implementation with $O(K)$ worst-case complexity and previous naive sorting-based implementation with $O(K \log K)$ worst-case complexity, 100 independent runs

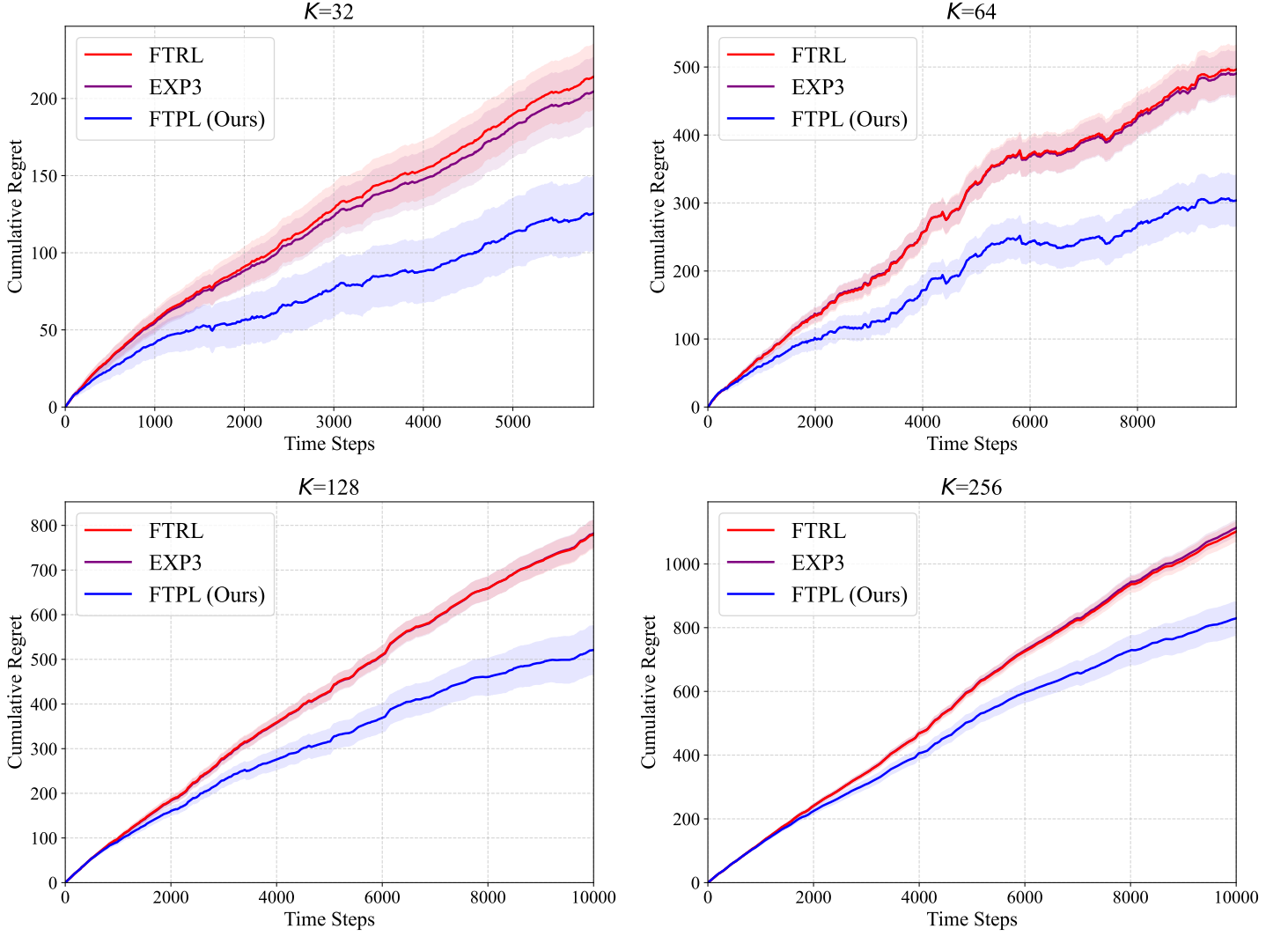


Figure B: Real-world data-based experiments, where we construct a robust offline oracle using the Movie-Lens 100K dataset to simulate online setting. 100 independent runs

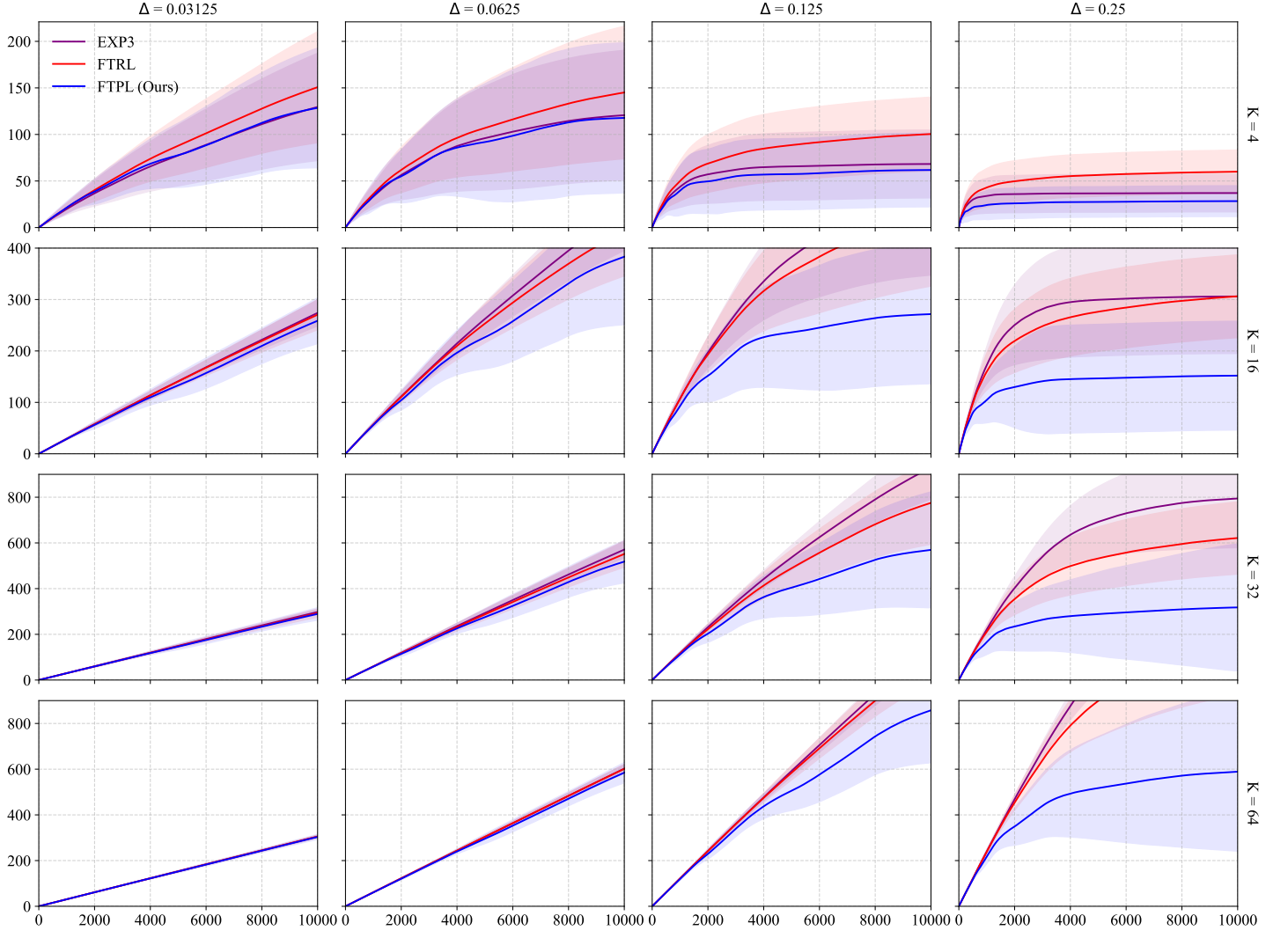
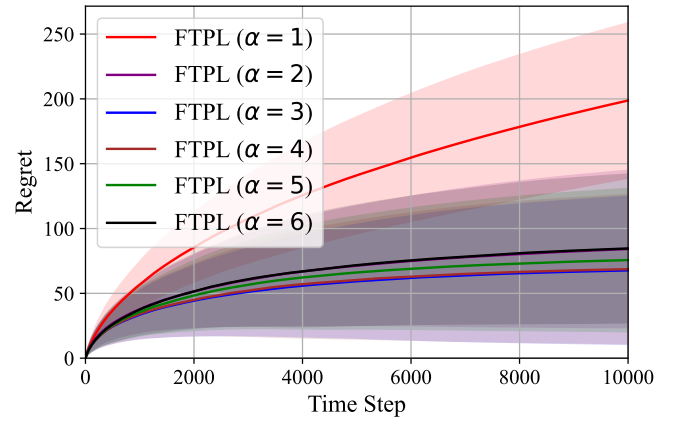
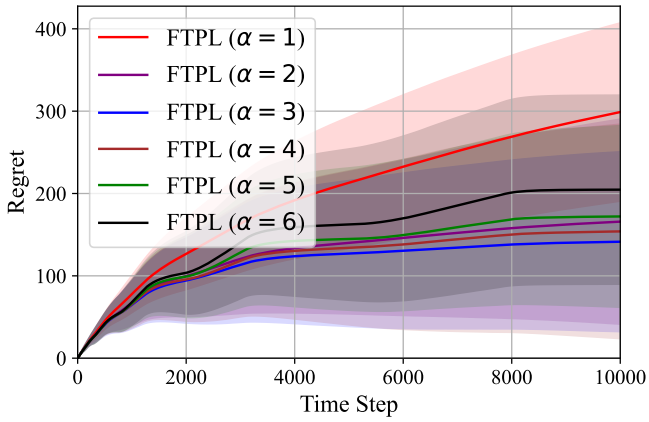


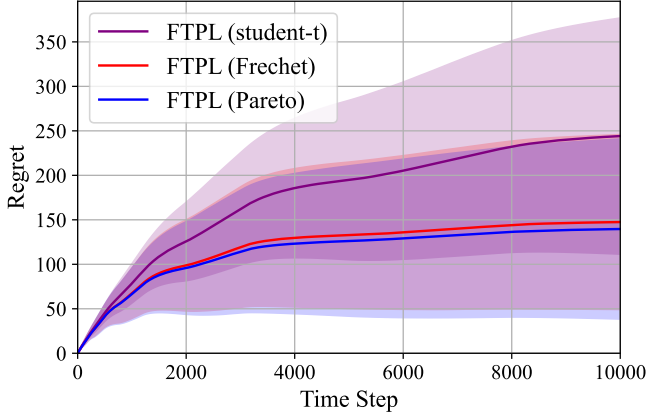
Figure C: Adversarial settings with various choices of the number of arms $K \in \{4, 16, 32, 64\}$ and the suboptimality gap $\Delta \in \{0.03125, 0.0625, 0.125, 0.25\}$ considered in Zimmert & Seldin (2021), 1000 independent runs



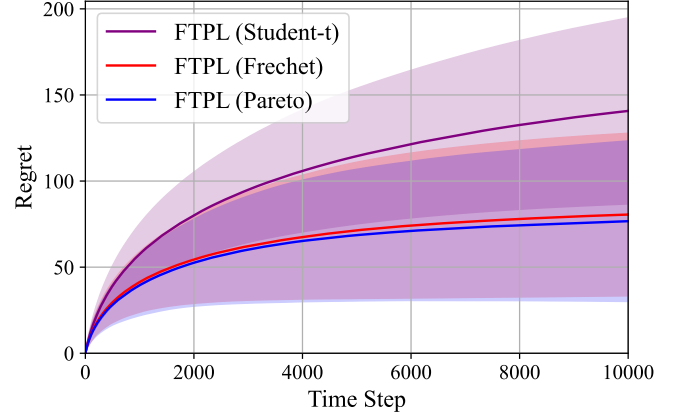
(i) Adversarial setting (that of Figure 2 in the submitted paper)

(ii) Stochastic setting (that of Figure 4 in the submitted paper)

Figure D: FTPL with different choices of α , 1000 independent runs

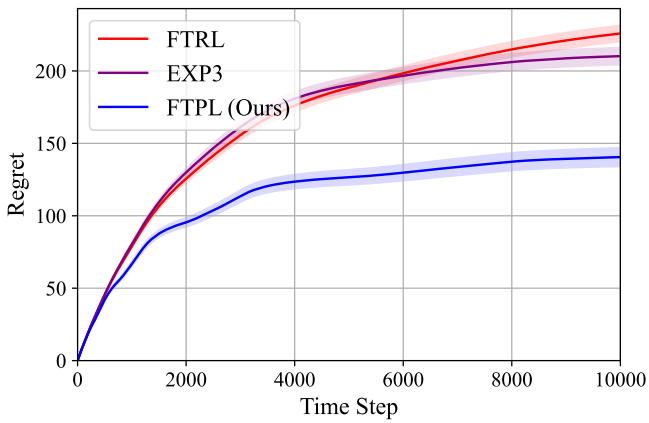


(i) Adversarial setting (that of Figure 2 in the submitted paper)

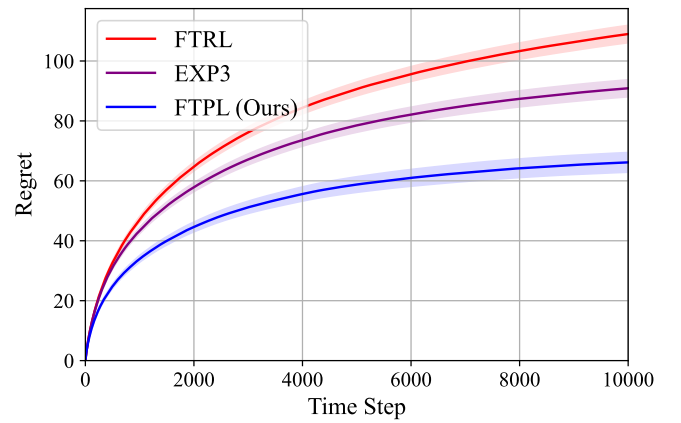


(ii) Stochastic setting (that of Figure 4 in the submitted paper)

Figure E: FTPL with different choices of perturbations with $\alpha = 3$, 1000 independent runs



(i) Figure 2 (adversarial setting)



(ii) Figure 4 (stochastic setting)

Figure F: Results with 95% confidence interval